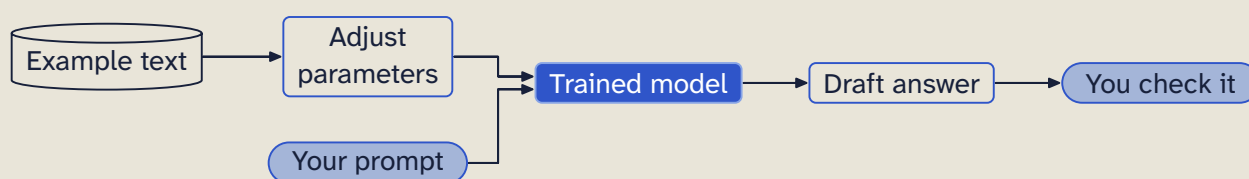


What is AI, really?

A plain-language guide to what today's AI does, how it learns, where it goes wrong, and how to start using it well.

From examples to answers

Top row: **training**, done once by the developer. Bottom row: what happens **each time you ask**.



The short answer

Artificial intelligence (AI) is a broad name for computer systems that do things we usually link with human thinking: recognising speech, translating, spotting patterns, writing. Most of what people call AI today is *machine learning*: instead of following rules a programmer wrote by hand, the system learns patterns from a very large number of examples.

How the learning works

Imagine building a spam filter. Rather than listing every spammy phrase, you show the system thousands of emails labelled “spam” or “not spam”. It adjusts millions of internal numbers, called *parameters*, until its guesses match the labels as often as possible. That adjustment is called *training*. Once trained, the same model can judge emails it has never seen.

What makes chatbots different

Large language models (LLMs), the technology behind chatbots such as ChatGPT, Claude and Gemini, are trained on a huge amount of text with one simple goal: predict the next piece of text. A piece is called a *token*, often a word or part of a word. To answer you, the model predicts one token, adds it to the conversation and repeats, thousands of times. Further training with human feedback teaches it to follow instructions and to decline some requests.

Why it can be wrong

An LLM produces the most plausible continuation, not a checked fact, so it can state false things fluently. These errors are often called *hallucinations*. Its knowledge stops at a training cut-off unless it is connected to search or to your documents, and it can repeat biases found in the text it learned from.

None of this makes the tool useless. It means the output is a draft for you to review.

What it is not

Today's AI systems do not understand the world the way people do, have no feelings and have no goals of their own. They are powerful pattern machines. Treat one like a fast, well-read assistant: excellent at first drafts, explanations and options, and in need of a human editor.

Glossary

Algorithm: a precise set of steps a computer follows.

Machine learning: learning patterns from examples instead of hand-written rules.

Model: the trained system that makes predictions.

Parameters: the internal numbers training adjusts; large models have billions.

Training data: the examples a model learns from.

Token: a chunk of text, often part of a word.

Prompt: the instructions and context you give a model.

Context window: how much text a model can consider at once.

Hallucination: a fluent answer that is not true.

Bias: unfair patterns a model picks up from its data.

Getting started well

Pick a task where a mistake is cheap: summarising a public article, brainstorming names for a project, or explaining a concept at your level. Give context, say what format you want, and ask follow-up questions. Check anything factual before you use it, and keep personal or confidential information out of tools your organisation has not approved.

Keep exploring

Elements of AI is a free online course for non-experts, created by the University of Helsinki and MinnaLearn. No maths or programming needed.

Scan the code or visit elementsofai.com



Try it in five minutes

1. Ask an AI tool to explain a topic you know well.
2. Spot one thing it got wrong or left out.
3. Ask it to fix that, and compare.

Before you trust an answer, ask:

1. Where could this be wrong?
2. How can I check it?
3. What happens if it is wrong?